

The contributions of root morphological characteristics and soil property to soil infiltration in a reseeded desert steppe

Bo Wang^{a,b,1}, Jing Liu^{a,b,1}, Zhigang Li^{a,b,*}, Stephen J. Morreale^c, Rebecca L. Schneider^c, Dongmei Xu^{a,b}, Xiongkui Lin^{a,b}

^a School of Agriculture, Ningxia University, Yinchuan, Ningxia 750021, China

^b Ningxia Grassland and Animal Husbandry Engineering Technology Research Centre, Yinchuan, Ningxia 750021, China

^c Department of Nature Resources and the Environment, Cornell University, Ithaca, NY 14886, USA

ARTICLE INFO

Keywords:

Soil infiltration
Root morphological characteristics
Soil porosity
Fine root
Reseeded desert steppe
Native plants

ABSTRACT

Infiltration of precipitation is the main source of water for soil water replenishment in arid and semiarid regions and plays a critical role in influencing the restoration of degraded grassland. However, the effects of reseeded vegetation on soil infiltration in grasslands remains unclear. Here, we measured soil infiltration rate (IR), root characteristics, and soil properties for four plant treatments (*Agropyron mongolicum*, *Lespedeza potaninii*, *A. mongolicum* × *L. potaninii* (1:1), and *A. mongolicum* × *Astragalus melilotoides* (1:1)) five years after reseeding and enclosure in degraded desert steppe, and continuously grazed steppe used as a control. Results showed that the IR among treatments presented as *A. mongolicum* > *L. potaninii* > *A. mongolicum* × *A. melilotoides* > *A. mongolicum* × *L. potaninii* > the grazed steppe. Correlation analysis indicated that root length density, root surface area, root volume, soil organic matter, non-capillary porosity, capillary porosity, and total porosity were positively correlated with initial, steady, and average IR ($p < 0.05$, or 0.01), yet root biomass and soil bulk density did not significantly influence IR ($p > 0.05$). The study also indicated that the higher IR of *A. mongolicum* was associated with higher soil organic matter and soil porosities at soil depths of 0–40 cm than in *L. potaninii* plots. Whereas, the higher IR of monocultures was associated with higher values of root length density, root surface area, and root volume at soil depths of 0–30 cm than mixtures. However, both root morphological characteristics and soil properties were significantly determined by the volumes of fine roots (diameter ≤ 2 mm). Therefore, in this study, the fine root volume played a crucial role in regulating soil infiltration via determining root morphological characteristics and forming soil pores and pathways.

1. Introduction

Grasslands are one of the main vegetation types in China that fulfill important functions and services in terrestrial ecosystems (Liu et al., 2011). However, grassland degradation has become a serious environmental problem in China due to climate variability and anthropogenic disturbances (Liu et al., 2014). Therefore, many practices such as reseeding, fertilization, irrigation, and enclosure have been adopted to restore degraded grasslands (Zhang et al., 2021).

Among restoration practices, reseeding is applied as one of the most common measures for grasslands (Butler and Haygarth, 2007; Necpálová et al., 2013). Particularly, reseeding native species not only can shorten restoration processes in degraded grasslands, but also can

accelerate the recovery of multifunctionality and maintain the stability of grassland ecosystems (Zhang et al., 2020). In China, reseeding native plant species to regenerate resilient and sustainable grassland ecosystems started as early as 1979 (Zhou et al., 2017). However, soil moisture is the key limiting factor that impacts both agricultural production and ecological restoration in arid and semiarid regions (Pachauri and Meyer, 2014; Guo et al., 2021). It's widely regarded that precipitation infiltration is the main source for soil water replenishment in arid and semiarid regions (Sun et al., 2018). Specifically, surface water infiltration is key to meeting plant water scarce regions, and also influences the success of vegetation restoration (Liu et al., 2020). Therefore, increases in surface soil water infiltration may be beneficial for the establishment and sustainability of reseeded grassland communities in arid and

* Corresponding author at: School of Agriculture, Ningxia University, No. 489, West Helan Mountain Road, Yinchuan, Ningxia 750021, China.

E-mail address: lizg001@sina.com (Z. Li).

¹ These authors contribute equally to this paper.

semiarid regions.

Many studies have demonstrated that soil water infiltration rate is generally influenced both by plant root systems (principally measured as root biomass, root length density, root surface area, and root volume) and soil properties (including soil porosity, bulk density, and soil organic matter) (Leung et al., 2015; Liu et al., 2015; Cui et al., 2019; Liu et al., 2020; Wu et al., 2021). However, among those factors, plant root systems are generally considered to have a greater potential for regulating soil water infiltration than soil properties (Jotisankasa and Sirirattanachai, 2017). Importantly, dead and decaying roots create a high density of macropores (or root channels) which can significantly increase both the infiltration rate (Wu et al., 2021) and subsequent preferential flow, resulting in rapid replenishing of soil water throughout the macropores (Guo et al., 2019). Live plant roots also can form microchannels, which play an important role in regulating soil water infiltration in arid and semiarid regions (Huang et al., 2017). In addition, plant root systems always input substantial carbon into the soil, which fosters the soil aggregate structure (Six et al. 2004; Huang et al., 2017). Fine roots, which turnover quickly (Solly et al., 2014), are more effective than coarse roots in increasing soil organic carbon and forming soil pores, and enhancing soil infiltration rate (Cui et al., 2019). It's also reported that high root biomass results in high soil water infiltration rate since high root biomass significantly increases soil organic matter and soil porosity (Huang et al., 2017; Liu et al., 2019). Furthermore, soil bulk density (Neris et al., 2012), initial soil water content (Cui et al., 2019), saturated hydraulic conductivity (Alaoui et al., 2011), and other soil properties (Wang et al., 2022) were also found to be the primary soil properties that influence soil infiltration. In summary, both root parameters and soil properties (especially root-induced soil properties) should be considered when evaluating the soil infiltration capacity of grasslands in arid and semiarid regions, especially in planning restoration efforts.

Due to their high nutrition value to livestock and significant restoration effect to degraded soil, both legumes and grasses are common plant taxa that have been widely reseeded in degraded grasslands in China (Zhang et al., 2020). Previous studies have also demonstrated that legumes and grasses have different capacities for regulating soil water infiltration due to their different root morphological characteristics and different contributions to soil property improvement (Huang et al., 2017; Cui et al., 2019; Liu et al., 2020). For instance, due to its greater root biomass and greater contribution in increasing soil organic matter, alfalfa (*Medicago sativa*) was shown to be better than grasses (*Agropyron cristatum*, *Stipa capillata*) in controlling soil water infiltration in a cultivated grassland in a semiarid region (Huang et al., 2017). Liu et al. (2020) also found that the leguminous species *Melilotus suaveolens* had stronger impact than gramineous species (*Panicum virgatum*, *Bromus inermis*, and *Miscanthus sinensis*) in regulating water infiltration in a cultivated grassland in an arid region, and root length density and root surface area were the main influencing factors. However, previous studies often considered the effect of monocultures on soil infiltration, yet, plants are seeded both in monocultures and mixtures in degraded grasslands (Zhang et al., 2020), and the monocultures and mixtures may differently influence soil infiltration. In order to fully understanding the effects of reseeding patterns on restoration of degraded grasslands, it's necessary to explore the effects of reseeded legumes and grasses on soil water infiltration, introduced as either monocultures, or in mixtures. In addition, though many previous studies investigated the soil infiltration in cultivated grasslands, most of the cultivated grasslands are planted with non-native forages, generally receiving irrigation and fertilization (Ghani et al., 2022; Kamran et al., 2022), and those measures may also influence soil water infiltration when compared with natural grasslands. Therefore, it's also necessary to investigate how reseeded native plants determine soil infiltration in natural grasslands under natural conditions.

In this study, we evaluated the soil infiltration rate of three native plant species including one native grass (*Agropyron mongolicum*) and two

legumes (*Lespedeza potaninii*, and *Astragalus melilotoides*) five years after being reseeded as monocultures or mixtures in a desert steppe. The overall goal of this study was to provide new insights into soil infiltration for vegetation restoration of degraded grasslands in semiarid regions of northern China. The specific objectives were to (1) compare soil infiltration rate among different reseeded plant communities; (2) determine the effects of root morphologic parameters and soil properties on soil infiltration; and (3) identify the main related factors that affect soil infiltration rate.

2. Materials and methods

2.1. Study site

The study site was located in Yanchi County, Ningxia (106° 30' – 107° 47' E, 37° 04' – 38° 10' N), northern China. The regional climate is a typical temperate continental and monsoonal climate. The average annual temperature is 8.1°C. The annual precipitation is 295 mm with ~ 60% of the precipitation occurring from June to October, and the average annual potential evapotranspiration is 2132 mm, with a frost-free period of about 162 d. The soil texture is primarily sandy loam. The natural vegetation of the site is desert steppe, and plant species mainly consist of *Agropyron mongolicum*, *Lespedeza potaninii*, *Astragalus melilotoides*, and others. All are contained within a long-term (≥10 years) enclosed grassland. An adjacent and unenclosed grassland has been degraded due to continuous heavy grazing by Tan sheep, and there the plant species are mainly *Artemisia scoparia*, *Corispermum mongolicum*, *Heteropappus altaicus*, *Salsola ruthenica* and others.

2.2. Seed preparation

Seeds of *A. mongolicum*, *L. potaninii* and *A. melilotoides* were collected in 2016 from the enclosed grassland at the study site. After collection, seeds were carefully cleaned, and the plump seeds were picked out for future use. The purities of *A. mongolicum*, *L. potaninii* and *A. melilotoides* were ~ 90%, 90%, and 85%, respectively; while the germination rates were ~ 76%, 85%, and 80%, respectively. Seeds were air-dried at room temperature and stored in paper bags for future reseeding.

2.3. Field experiment design

Five treatments were set up including desert steppe with two monocultures of (a) *A. mongolicum* and (b) *L. potaninii* alone; two treatments with mixtures of (c) *A. mongolicum* × *L. potaninii* (1:1) and (d) *A. mongolicum* × *A. melilotoides* (1:1); and a fifth treatment (e) an adjacent continuously grazed steppe selected as a Control (Fig. 1). The above reseeding patterns mimicked the native vegetation communities of a 10-year enclosed steppe within in the study site. In each treatment, there were three replicate plots (5 m × 10 m) laid out in a stratified random pattern. Reseeded plots were separated from each other by 2-m spacing between plots and 5-m alleys between rows. The reseeded plots were initially tilled to a depth of ~ 10 cm before starting the experiment. In the grazed steppe, three plots were randomly selected, with grazing density of 1–2 Tan sheep hm², located ~ 50 m away from the reseeded plots. The grazing steppe without reseeding and enclosure management.

Our reseeding experiment was established in July 2017. After reseeding, the plots were enclosed and plant growth completely depended on normal rainfall without additional fertilization and no animal disturbance. The plant growth parameters including plant height, coverage, density, and above-ground biomass for each treatment are shown in Table S1.

2.4. Measurement of soil infiltration rate

The soil infiltration rates were measured in August 2022 by a double-ring infiltrometer (Liu et al., 2020; Cui et al., 2021). The infiltrometer

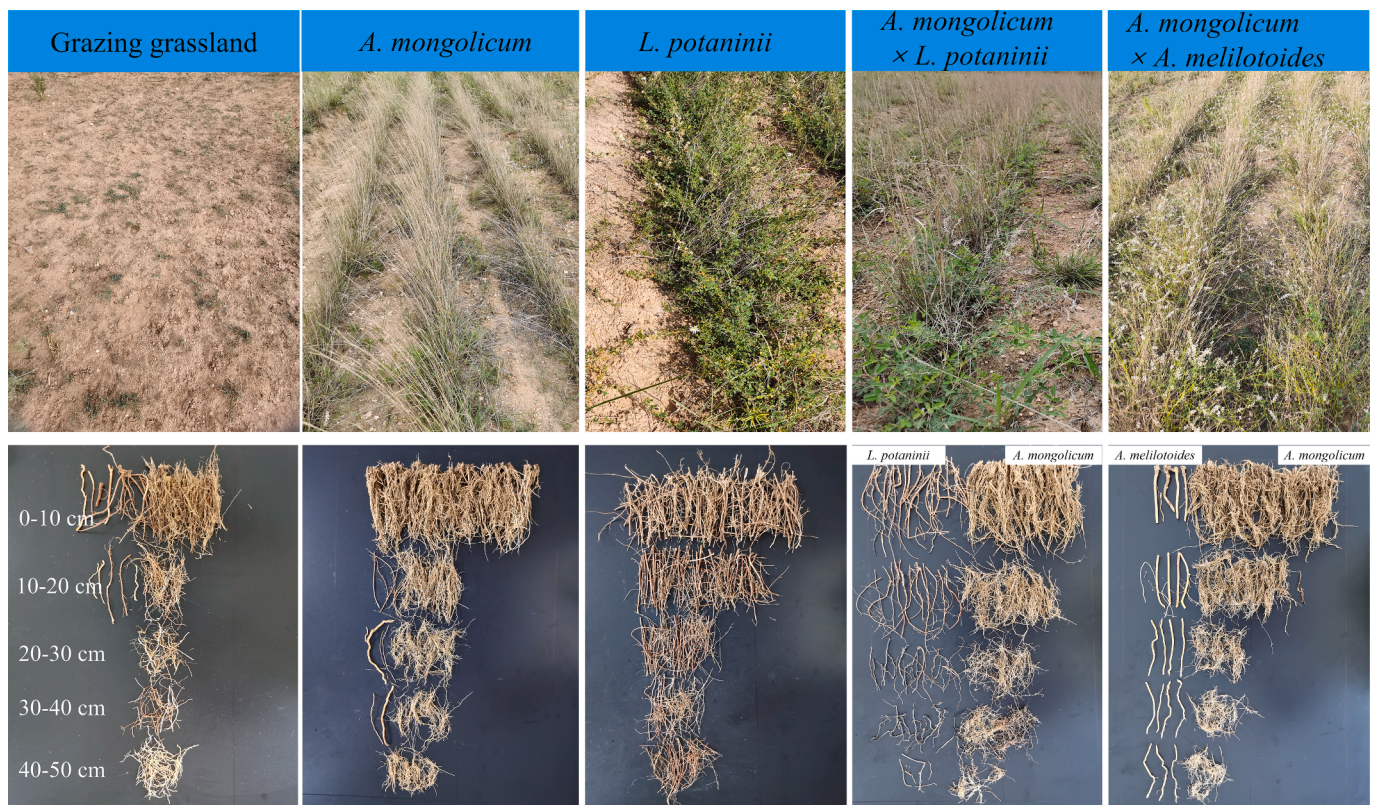


Fig. 1. Schematic showing experimental design (top figures) and root distribution at different soil depths (bottom figures).

consisted of an inner ring with a diameter of 32 cm and an outer ring with a diameter of 60 cm. Three flat areas were chosen for three replicate measurements in each plot. At harvest, all plants were cut off at ground level and the litter layer was carefully removed before measurement. The double-ring infiltrometer was gently and vertically inserted into the soil to a depth of ~ 5 cm. Meanwhile, the soil outside the ring was compacted to avoid water leaking out of the ring. Afterwards, water was added into the inner and outer rings respectively until the water depth reached 5 cm at the same time. The time for the water level of the inner ring to drop by 5 mm was recorded during the experiment. The inner and outer ring water were then refilled to a height of 5 cm when the water line dropped to 1 cm. The water levels of the inner and outer rings were consistent during the infiltration process. The first three minutes of the infiltration process were calculated as the initial infiltration rate (IIR), the average of the last three fairly constant infiltration rates was calculated as the steady-state infiltration rate (SIR), and the average of the infiltration process for the entire measurement period was recorded as the average infiltration rate (AIR) (Liu et al., 2019, 2020).

2.5. Measurement of plant root characteristics

After infiltration measurements were recorded, vertical soil profiles within 0.2 m × 0.2 m quadrats were excavated at the same three points in each plot. Soil samples were collected to a depth of 50 cm with five intervals including 0–10, 10–20, 20–30, 30–40 and 40–50 cm in each plot. The three replicate soil samples were mixed to create a single sample for each plot, and then quickly transported to the laboratory. The roots were separated from the soil sample by using a 5 mm mesh sieve. After separation, the roots were then placed in a basin and gently washed with water to remove soil and debris. The roots were scanned and analyzed using the WinRHIZO (Pro 230e) root analysis system for morphological parameters including root length density (RLD) (mm (100 cm³)⁻¹), root surface area (RSA; cm²) and root volume (RV; cm³)

(Liu et al., 2020). After the above measurements, the separated roots were dried at 75 °C for 48 h, and then weighed as root biomass (RB).

2.6. Measurements of soil properties

Soil organic matter was measured by the K₂Cr₂O₇–H₂SO₄ oxidation method (Nelson and Sommers, 1982). Soil bulk density (BD), total porosity (TP), capillary porosity (CP), and non-capillary porosity (NCP) were determined according to Liu et al. (2009). Briefly, a metal ring sampler, 5.0 cm diameter × 5.0 cm height, was used to determine total and capillary porosity based on soil water retention characteristics. Capillary porosity was determined based on the measured volume of water retained after placing the soil core in a tray with a 5-mm level of water until a filter paper at the top of each core became moist. Total porosity was based on the measured volume of water retained after the soil core had been placed in a 5-cm deep tray filled with water for 24 h. Non-capillary porosity was calculated as the difference between total and capillary porosity. Bulk density was determined by drying a soil core under 105 °C for 48 h, and then calculated based on the dry soil weight and metal ring volume.

2.7. Statistical analysis

One-way analysis of variance (ANOVA) tests was used to compare plant growth parameters, infiltration rates, root morphological parameters, and soil porosities among treatments. Tukey HSD tests were used to make *post hoc* multiple pairwise comparisons among all treatments, when data exhibited homogeneity of variance. Non-linear regression models were used to analyze the relationship between infiltration time and infiltration rates, as well as the relationships between fine root volume and root length density, root surface area, total root volume, and root biomass. Pearson's correlation analysis was applied to reveal relationships between soil water infiltration rates and root morphological parameters, soil water infiltration rates and soil properties, and soil

properties and root biomass and root volume. Stepwise multiple regression analysis was used in determining the primary influence factors of root parameters and soil properties on soil infiltration rates. All above data analyses were conducted by IBM SPSS Statistics 20.0, and a p -value ≤ 0.05 was considered significant.

3. Results

3.1. Soil water infiltration rates under the different reseeded plants

The soil infiltration rates of different treatments decreased with increased infiltration time, and the changes of infiltration rates with time for all treatments were best displayed as robust power function models ($R^2 = 0.857\text{--}0.917$, $p < 0.001$; Fig. 2a). The infiltration rates among treatments decreased in the following order: *A. mongolicum* > *L. potaninii* > *A. mongolicum* $\times$ *A. melilotoides* > *A. mongolicum* $\times$ *L. potaninii* > grazed unseeded grazed steppe calculated for the entire measurement period (Fig. 2a). The initial infiltration rate (IIR), steady infiltration rate (SIR), and average infiltration rate (AIR) also followed the same pattern among treatments ($p < 0.05$; Fig. 2b–2d), except there was no difference between *A. mongolicum* $\times$ *A. melilotoides* and unseeded grazed steppe for IIR ($p > 0.05$; Fig. 2b).

3.2. Root distribution and root morphological characteristics under the different reseeded plant species

Based on root distribution patterns, all reseeded treatments exhibited

stronger root systems than that of the unseeded grazed steppe (Fig. 1). The grass species, *A. mongolicum*, developed a stronger root system than the two legumes in their mixtures, though it was initially planted with *L. potaninii* and *A. melilotoides* in a 1:1 ratio at the experiment start.

Further analysis demonstrated that root length density (RLD), root surface area (RSA), root volume (RV) and root biomass (RB) decreased with increasing soil depth for all treatments (Fig. 3). Average RV and average RB at the 0–30 cm soil depth accounted for 94.6% of total RV and 93.3% of total RB, respectively. A one-way ANOVA indicated that the four reseeded treatments had higher values in RLD, RSA, RV, and RB than the unseeded grazed steppe at each soil depth interval ($p < 0.05$; Fig. 3). As a whole for the soil depth intervals between 0 and 30 cm, the RLD, RSA, and RV in the two monocultures were higher than the two mixtures, but there were no differences between the two monocultures (except for RLD, RSA, and RV of *L. potaninii* which were higher than *A. mongolicum* at soil depth of 20–30 cm ($p < 0.05$)) nor between the two mixtures. Conversely, *A. mongolicum* together with *A. melilotoides* $\times$ *L. potaninii* presented higher RB values than the other two reseeded treatments.

We also analyzed the volume distributions of the different root diameters (Fig. 4). Overall, reseeded *A. mongolicum*, *A. mongolicum* $\times$ *A. melilotoides* and *A. mongolicum* $\times$ *L. potaninii* increased the volume of their fine roots (i.e. diameter of 0–2 mm) at the 0–30 cm soil depth when compared with the unseeded grazed steppe, yet there were no differences among them for the entire soil depth ($p > 0.05$), and there was no difference between *L. potaninii* and unseeded grazed steppe for entire soil depths as well ($p > 0.05$). In contrast, only reseeded *L. potaninii*

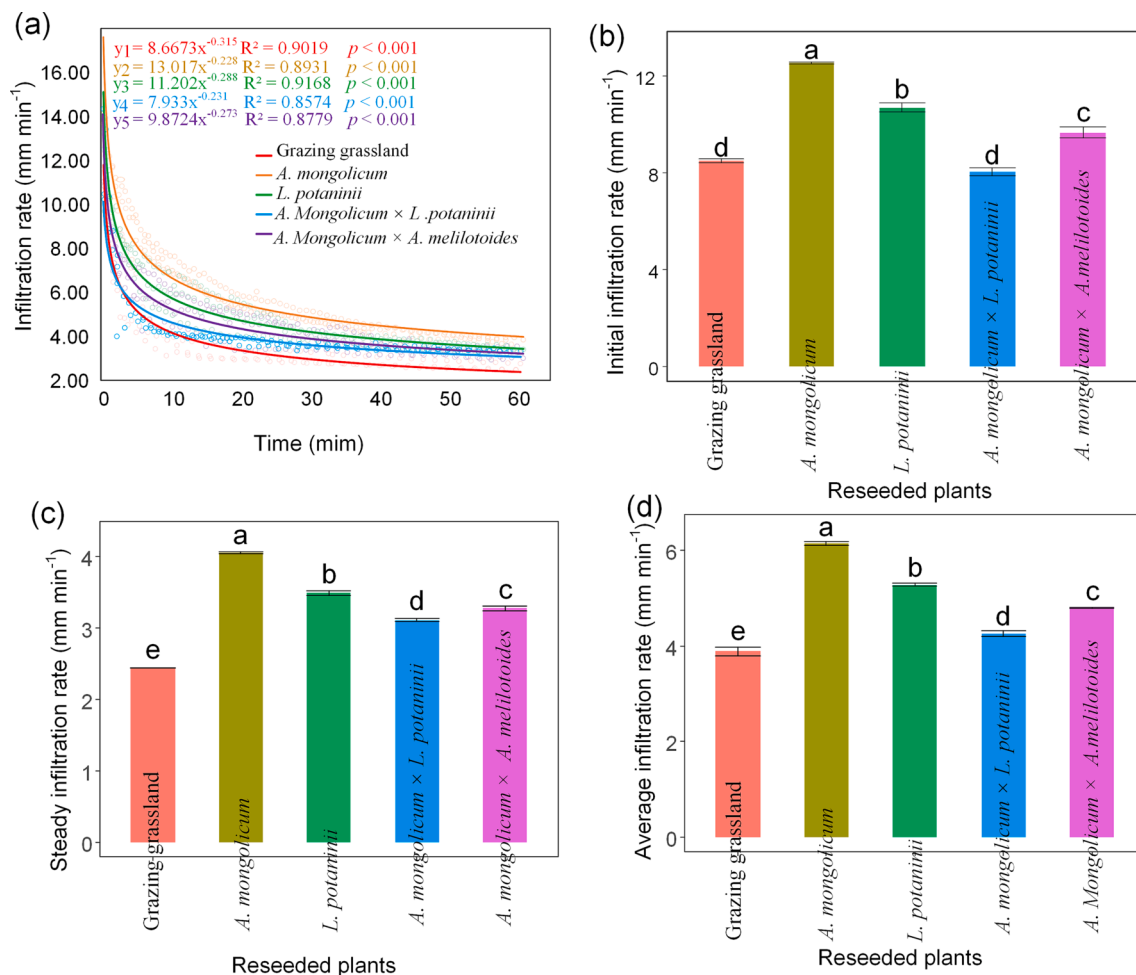


Fig. 2. changes of soil infiltration rate with time (a), as well as initial infiltration (b), steady infiltration (c), and average infiltration (d) in different reseeded plant communities. Values are means $\pm$ SE ($n = 3$). Different letters are significantly different at the 0.05 level.

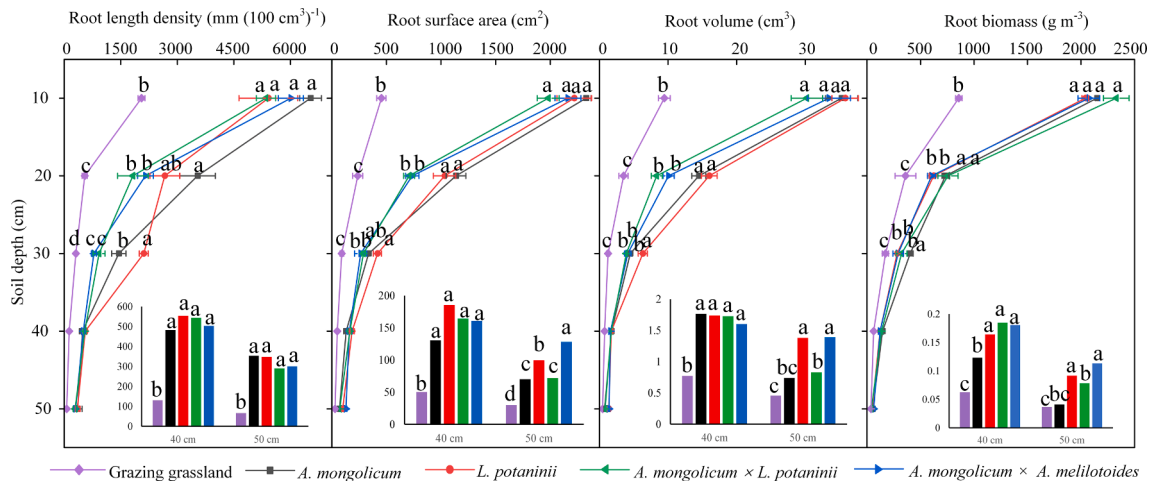


Fig. 3. Changes in root characteristics of different reseeded plant communities at different soil depths. Values are means $\pm$ SE ($n = 3$). Different letters are significantly different at the 0.05 level at each soil depth.

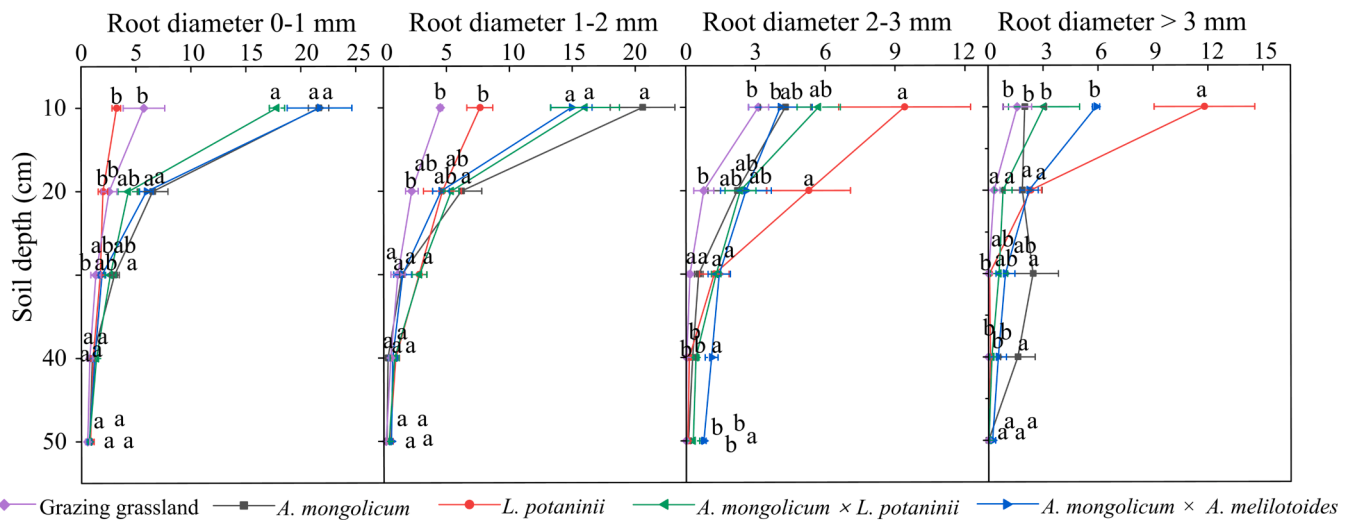


Fig. 4. The volume distributions of different diameters of root for reseeded plant communities. Values are means $\pm$ SE ($n = 3$). Different letters are significantly different at the 0.05 level at each soil depth.

increased the volumes of roots with diameter ≥ 2 mm at soil depth of 0–20 cm when compared to grazing steppe ($p < 0.05$). In addition, regression analysis showed that RLD, RSA, RV, and RB were significantly increased as fine root volumes increased ($R^2 = 0.599\text{--}0.903$, $p < 0.001$; Fig. 5).

3.3. Soil properties under the different reseeded plant species

As a whole, the values of the soil organic matter (SOM), non-capillary porosity (NCP), capillary porosity (CP), and total porosity (TP) in all reseeded treatments were higher than unseeded grazed steppe at soil depths of 0–40 cm. Among the reseeded treatments, *A. mongolicum* exhibited the highest values, but the other treatments exhibited no consistent patterns (Fig. 6). There were no differences in SOM, NCP, CP, or TP among all treatments at soil depths of 40–50 cm ($p > 0.05$; Fig. 6). In contrast, all reseeded treatments exhibited lower BD values than unseeded grazed steppe at soil depths of 0–40 cm, with *A. mongolicum* exhibiting lowest BD values at soil depth of 0–20 cm, and *A. mongolicum* $\times$ *A. melilotoides* and *A. mongolicum* $\times$ *L. potaninii* exhibited lowest values at soil depths of 20–30 cm and 30–40 cm, respectively. There were no differences for BD among all treatments at soil depths of 40–50 cm ($p > 0.05$; Fig. 6).

Correlation analysis indicated that soil properties (except SOM at 30–40 cm) were not correlated with RB across all soil depths ($p > 0.05$), total RV was positively correlated with NCP at soil depths of 0–10 cm and 30–40 cm, and also positively correlated with CP at soil depths of 10–20 cm ($p < 0.05$; Table 1). However, as a whole, most soil property indices in the shallow soil depths of 0–20 cm were correlated with fine root volumes ($p < 0.05$), whereas most soil property indices at deeper soil depths of 30–50 cm were correlated with root volume for coarser root diameters i.e. > 2 mm ($p < 0.05$; Table 1).

3.4. Relationships of soil infiltration rate with root characteristics and soil properties

Overall, correlation analysis indicated that all three measures of infiltration rate (IIR, SIR, and AIR) were primarily positively correlated with root properties and soil properties at soil depths of 0–30 cm (Table 2). Specifically, IIR, SIR, and AIR were positively correlated with RLD, RSA, and RV at 0–30 cm soil depths ($p < 0.05$ or 0.01), while both SIR and AIR were only positively correlated with RV at 30–40 cm soil depths ($p < 0.05$) and RLD at 40–50 cm soil depths ($p < 0.05$). In addition, IIR, SIR, and AIR were also positively correlated with NCP and TP at 0–20 cm soil depths ($p < 0.05$ or 0.01) and were positively

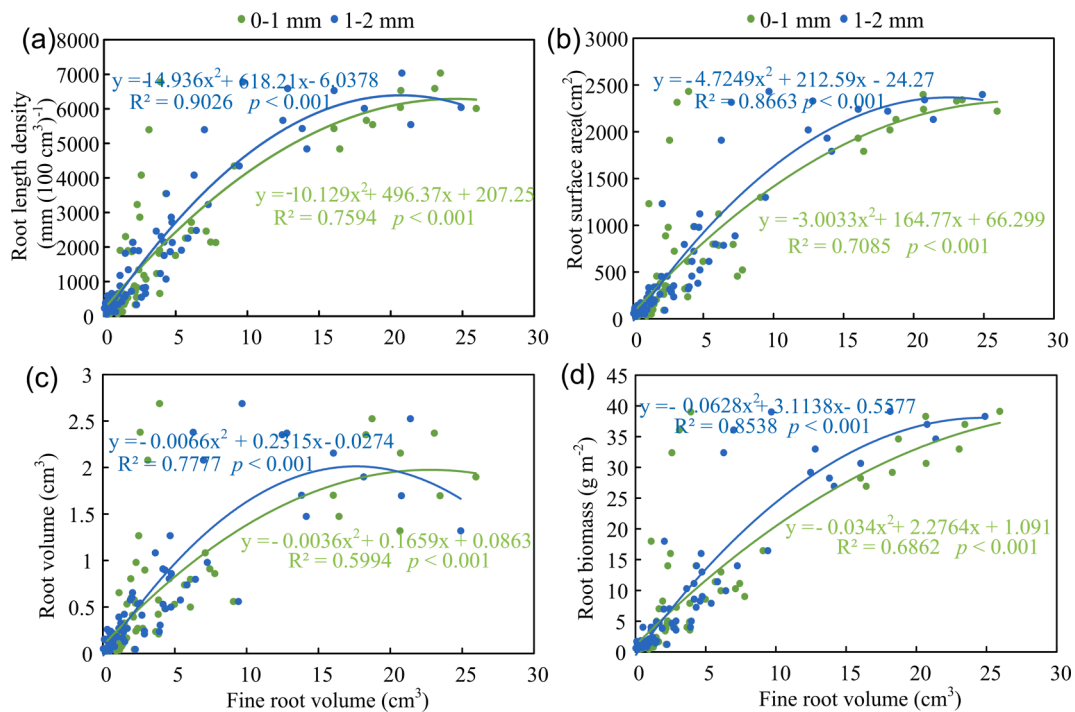


Fig. 5. The correlations between fine root (diameter with 0–1 mm and 1–2 mm) volume and root length density (a), root surface area (b), total root volume (c), and total root biomass (d).

correlated with TP at 10–30 cm soil depths ($p < 0.05$; except IIR at 10–20 cm soil depth). Moreover, IIR, SIR, and AIR were also positively correlated with SOM at 30–40 cm soil depths ($p < 0.01$), yet SIR was positively correlated with SOM at 40–50 cm soil depths ($p < 0.01$), and AIR was positively correlated with CP at 30–40 cm soil depths ($p < 0.05$). Notably, the stepwise multiple regressions indicated that the most important factor influencing IIR and AIR was the root surface area (RSA) at soil depths of 10–20 cm, while the most important factors influencing SIR were RSA at soil depths of 10–20 cm and the root length density (RLD) at soil depths of 0–10 cm (Table 3).

4. Discussion

Though many studies had evaluated the effects of reseeded vegetation on soil nutrients (Ji et al., 2022), soil physical property (Li et al., 2022), soil microbial community (Zhou et al., 2019), and vegetation (Tian et al., 2022) in grasslands, yet there are few studies concern its effects on soil infiltration, even if soil water is regarded as the key limiting factor that impacts ecological restoration in arid and semiarid regions (Pachauri and Meyer, 2014; Guo et al., 2021). Interestingly, our study revealed that all reseeded native plants significantly increased soil infiltration rate when compared with unseeded grazed desert steppe, and initial infiltration rate, steady infiltration rate, and average infiltration rate varied in a consistent pattern among the five treatments. Initial infiltration rate usually has the most critical effects on the runoff process at early rainfall stages, and steady infiltration rate reflects soil infiltrability and soil infiltration at later rainfall stages (Huang et al., 2019), while average infiltration rate was used to evaluate the overall infiltration rate. Therefore, our results indicated that those reseeded native plants have positive effects in restoring degraded desert steppe since reseeding measures can decrease runoff and capture rainfall during the entire rain period.

Our study also found that soil water infiltration processes in the desert steppe were significantly influenced both by root morphological characteristics and soil properties. Similarly, previous studies also found that soil water infiltration processes were influenced both by root and soil properties, and that soil properties (soil organic matter, soil bulk

density, or soil porosities) played comparable roles with as root properties in controlling soil infiltration (Huang et al., 2017; Cui et al., 2019; Liu et al., 2019, 2020; Shi et al., 2022). However, our study indicated that the root surface area and the root length density in the surface soil played a key role in determining soil infiltration based on the stepwise multiple regressions. The root length density reflects the extent of the distribution, interpenetration and interweaving of the root system in the soil (Liu et al., 2020), while root surface area directly reflects the developed extent of root channel in soil. Hence, both root length density and root surface area are pivotal factors affecting soil infiltration. Specifically in our study, more than 90% of roots were distributed in the surface 0–20 cm depths, which also resulted in high values of root length density and root surface area in these shallow depths and ultimately determined water infiltration as other studies reported (Hao et al., 2020; Liu et al., 2020). Moreover, plant roots near the surface soil also can improve soil porosity, which also benefits water infiltration (Leung et al., 2015; Huang et al., 2017). The further analysis of this study revealed that both the root surface area and root length density were significantly affected by the volume of fine roots of diameter ≤ 2 mm. Thus, higher root surface area and root length density of fine roots translates to more pores and pathways which allow more water to flow into and through the soil (Wang et al., 2022). Some other studies even found that fine root length density was the best predictor for soil properties and infiltration (Hao et al., 2020). Similarly, Luo et al. (2019) also found that fine roots facilitate preferential flow in the surface 0–40 cm of the forest soil layer, likely through the quick turnover of fine roots, which results in abundant preferential flow paths (Zhang et al., 2015).

Aside from the direct contribution of roots to soil physical properties, other studies also indicate that long-term natural grassland restoration (Zhao et al., 2013) or cultivated grassland establishment (Huang et al., 2019) accumulates organic matter, which helps improve soil structural properties, thus significantly enhancing soil infiltration capacity. Particularly, root-induced increases in soil organic matter and soil aggregates significantly influence soil infiltrability (Hao et al., 2020). For instance, high root biomass was proved can significantly increase soil organic matter and soil porosity and decreases bulk density, which ultimately contributes to high soil water infiltration rate (Huang et al.,

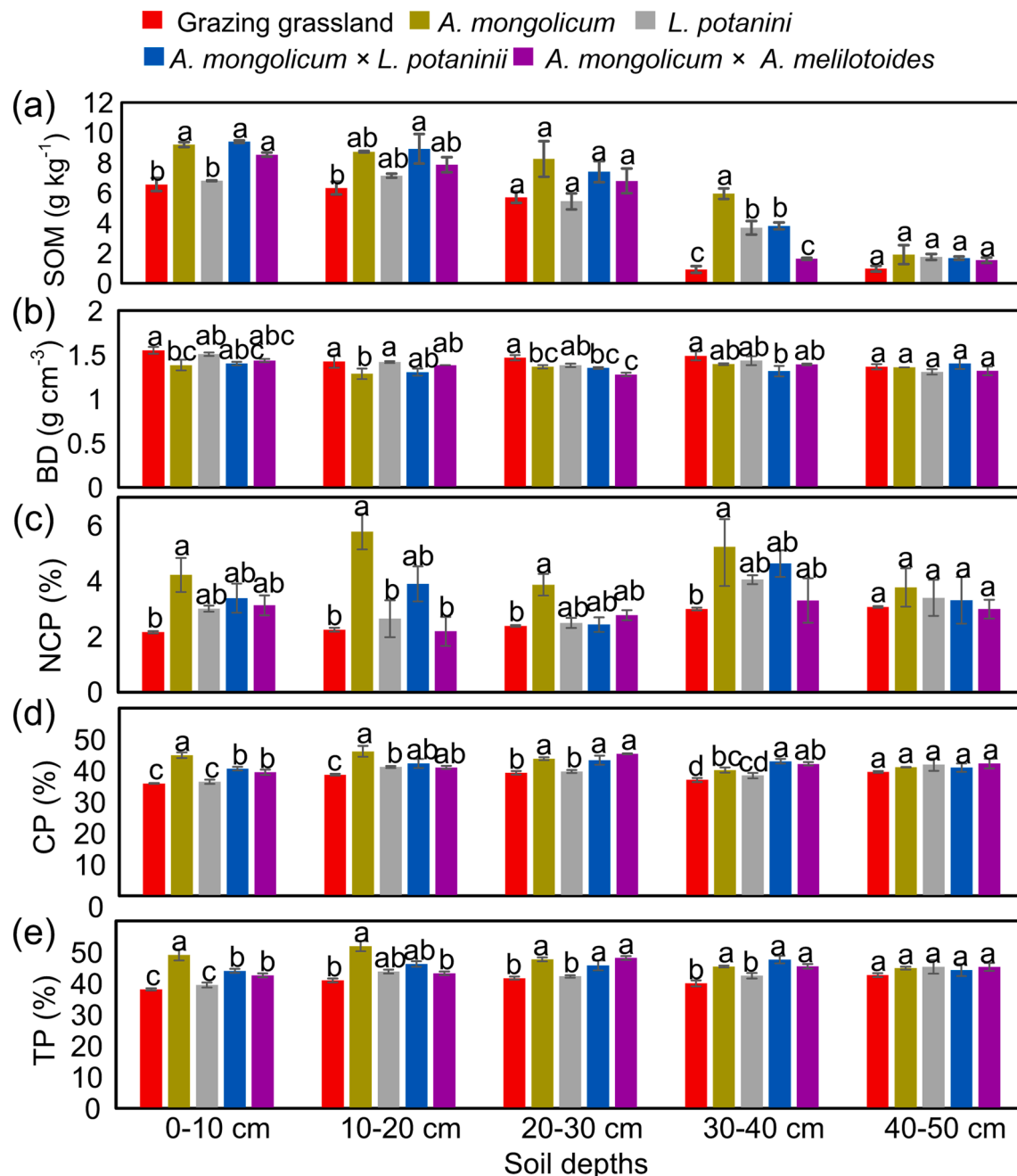


Fig. 6. Soil properties including bulk density (a), no capillary porosity (b), capillary porosity (c), and total porosity (d) at different soil depths in different reseeded plants. BD, bulk density; NCP, no capillary porosity; CP, capillary porosity; TP, total porosity. Values are means $\pm$ SE ($n = 3$). Different letters are significantly different at the 0.05 level at each soil depth.

2017; Liu et al., 2019). However, though our study also showed that reseeded treatments increased root biomass, yet the root biomass had no strikingly contribution to improve soil organic matter, soil bulk density, and soil porosities, and ultimately total root biomass as well as soil bulk density had no contribution to soil infiltration rates at all tested soil depths. We assumed that the differences among these studies may be due to the differences in plant species and/or soil properties. For instance, previous studies found leguminous species (*Medicago sativa*, *Melilotus suaveolens*) have greater root biomass and root volume than grasses, which also resulted in greater infiltration in leguminous grassland than gramineous grassland (Huang et al., 2017; Liu et al., 2020). In contrast, our study found that gramineous species contributed higher root biomass and soil infiltration than leguminous species in the reseeded steppe. Moreover, the soil in our study was also sandier than those

previous studies, and this very sandy soil may also weaken the effect of soil bulk density on infiltration in our study.

Of particular note, we found that the increased fine root volumes significantly improved other soil properties such as soil organic matter, non-capillary porosity, capillary porosity, and total porosity, and ultimately enhanced the soil water infiltration. This finding corroborated the report from Cui et al. (2019) who also found that the percentage of fine root volume was positively correlated with infiltration rate, because those fine roots could form substantial soil pores. Hao et al. (2020) also found that fine roots played critical roles in mediating aggregate development and porosity, compared to coarse root-induced macropores. In summary, the fine root volume played a critical role in regulating soil infiltration in this study, since it significantly determined root morphological characteristics (RLD, RSA, and RB) and soil properties

Table 1
Correlation coefficients between soil properties and root biomass and root volume.

Soil depths	Soil properties	RV of 0–1 mm	RV of 1–2 mm	RV of 2–3 mm	RV of > 3 mm	Total RV	RB
0–10 cm	SOM	0.817**	0.858**	−0.134	−0.328	0.483	0.253
	BD	−0.544*	−0.480	−0.150	−0.168	−0.107	−0.374
	NCP	0.443	0.589*	0.414	0.068	0.650**	0.189
	CP	0.671**	0.761**	0.036	−0.204	0.193	0.086
	TP	0.732**	0.825**	0.093	−0.225	0.321	0.129
10–20 cm	SOM	0.556*	0.486	0.070	−0.100	0.425	0.074
	BD	−0.436	−0.468	−0.201	0.048	−0.106	−0.065
	NCP	0.166	0.381	0.324	0.052	0.229	−0.052
	CP	0.593*	0.725**	0.164	0.221	0.604*	0.011
	TP	0.470	0.749**	0.341	0.227	0.513	0.136
20–30 cm	SOM	0.577*	0.099	0.030	0.270	0.395	−0.142
	BD	−0.183	−0.170	−0.491	−0.572*	−0.304	−0.295
	NCP	0.254	−0.246	−0.021	0.366	−0.007	0.150
	CP	0.425	0.125	0.375	0.654**	0.032	0.018
	TP	0.486	0.004	0.286	0.724**	0.036	0.068
30–40 cm	SOM	0.504	0.220	0.455	−0.036	0.381	0.553*
	BD	−0.475	0.068	−0.396	−0.387	−0.279	−0.227
	NCP	0.707**	0.393	0.193	0.387	0.564*	0.116
	CP	0.393	−0.064	0.776**	0.361	0.416	0.457
	TP	0.554*	−0.036	0.718**	0.480	0.418	0.368
40–50 cm	SOM	0.427	0.108	−0.002	−0.157	0.194	0.158
	BD	0.373	0.075	0.117	−0.206	−0.215	−0.187
	NCP	0.186	0.057	0.040	−0.035	−0.179	−0.090
	CP	−0.025	0.518*	0.473	0.647**	0.168	0.099
	TP	0.018	0.511	0.462	0.614*	0.161	0.088

RB, root biomass; RV, root volume; 0–1 mm, 1–2 mm, 2–3 mm, and > 3 mm are diameters of root; SOM, soil organic matter; BD, bulk density, NCP, non-capillary porosity; CP, capillary porosity; TP, total porosity. * $p < 0.05$, ** $p < 0.01$.

Table 2
Correlation coefficients between plant root characteristics and infiltration rates, as well as between plant root characteristics and soil properties.

Soil depths	Infiltration rates	Root parameters				Soil properties				
		RLD	RSA	RV	RB	SOM	BD	NCP	CP	TP
0–10 cm	IIR	0.562*	0.550*	0.537*	0.209	0.142	−0.256	0.543*	0.338	0.556*
	SIR	0.829**	0.851**	0.829**	0.502	0.511	−0.350	0.629*	0.419	0.653**
	AIR	0.715**	0.724**	0.705**	0.373	0.332	−0.498	0.715**	0.473	0.742**
10–20 cm	IIR	0.629*	0.779**	0.756**	0.039	0.171	−0.210	0.603*	0.432	0.626*
	SIR	0.796**	0.918**	0.897**	0.234	0.496	−0.324	0.693**	0.539*	0.738**
	AIR	0.825**	0.893**	0.860**	0.137	0.348	−0.441	0.769**	0.549*	0.797**
20–30 cm	IIR	0.593*	0.568*	0.541*	0.211	0.126	−0.185	0.157	0.627*	0.313
	SIR	0.764**	0.746**	0.760**	0.431	0.309	−0.261	0.281	0.567*	0.405
	AIR	0.779**	0.707**	0.681**	0.304	0.276	−0.398	0.423	0.527*	0.519*
30–40 cm	IIR	0.139	0.132	0.307	−0.045	0.678**	−0.079	−0.123	0.367	0.042
	SIR	0.375	0.379	0.514*	0.283	0.849**	−0.069	0.049	0.507	0.242
	AIR	0.454	0.407	0.561*	0.135	0.786**	−0.267	0.286	0.552*	0.459
40–50 cm	IIR	0.332	0.229	0.271	−0.069	0.406	−0.263	0.137	0.303	0.220
	SIR	0.568*	0.371	0.418	0.058	0.538*	−0.259	0.233	0.329	0.315
	AIR	0.564*	0.404	0.461	0.000	0.468	−0.203	0.284	0.308	0.353

IIR, initial infiltration rate; SIR, steady infiltration rate; AIR, average infiltration rate; RLD, root length density; RSA, root surface area; RV, root volume; RB, root biomass; SOM; soil organic matter; BD, bulk density, NCP, non-capillary porosity; CP, capillary porosity; TP, total porosity. * $p < 0.05$, ** $p < 0.01$.

Table 3
Stepwise multiple regressions between infiltration rates (AIR, IIR and SIR) and root characteristics and soil properties.

Infiltration rates	Regression equations	R ²	p
IIR	$Y = 0.769\text{RSA}_{20} + 0.964$	0.559	< 0.001
SIR	$Y = 0.327\text{RLD}_{10} + 0.676 + \text{RSA}_{20} + 1.923$	0.874	< 0.001
AIR	$Y = 0.860\text{RSA}_{20} + 3.285$	0.721	< 0.001

IIR, the initial infiltration rate; SIR, the steady infiltration rate; AIR, the average infiltration rate; RSA_{20} , root surface area at the depths of 10–20 cm; RLD_{10} , root length density at the depths of 0–10 cm.

(SOM, NCP, CP, and TP).

A recent study demonstrated that different plant communities disparately affect soil infiltration in a semiarid desert steppe, due to their varied root systems and soil properties among plant communities (Shi et al., 2022). We also noticed that the soil infiltration properties differed

among reseeded plant communities in this study. However, we found there were only slight differences in root length density, root surface area, root volume, and root biomass between two monocultures, but the *A. mongolicum* treatment resulted in higher infiltration rates than the *L. potanini*. Further analysis revealed the volumes of fine root (diameter ≤ 2 mm) of *A. mongolicum* were significantly higher than that of *L. potaninii*, which may contribute to the greater infiltration rate, as was also documented by Cui et al. (2019). In other words, the difference of soil water infiltration between two monocultures was mainly determined by soil organic matter and soil porosities via varied fine root volumes, since finer roots exhibit substantially higher decomposition rate (Freschet and Roumet, 2017), resulting in higher level of root exudates and organic input (Galloway et al., 2018; Le Bissonnais et al., 2018). All those contributions form fine roots ultimately enhanced aggregate stability and soil infiltration. Soil organic matter and porosities in *A. mongolicum* were also observed higher than *L. potaninii* at soil depths of 0–40 cm in this study. Similarly, Hao et al. (2020) also found

that the infiltration rates of two planted grasses were higher than two planted leguminous shrubs, and the most likely driver was the higher amounts of fine roots of two grasses, which significantly increased soil organic matter, aggregate stability, and porosities.

Liu et al. (2019) reported that leguminous monocultures had higher infiltration rates than leguminous-gramineous mixtures in a cultivated grassland, and found initial soil moisture and root biomass were the main factors influencing infiltration rate. Though we did not measure initial soil moisture, we found root length density, root surface area, and root volume played important role in determining soil infiltration. And notably, this study showed that the values of root length density, root surface area, and root volume of monocultures were higher than that of mixtures. Especially, though *A. mongolicum* was a mixture of seeds with *L. potaninii* and *A. melilotoides* at a ratio of 1:1 at the experiment start, the grass species developed more robust root systems than two legumes and consequently occupied a greater percentage of root volume in the mixed root systems. Liu et al. (2014) also reported that gramineous species exhibited strong competition for a legume, and the legume was greatly suppressed when reseeded as a mixture in a semiarid grassland. Furthermore, the reseeded *A. mongolicum* was also observed to have higher fine root volume than two leguminous species. Hence, those results suggest *A. mongolicum* may play an important role in regulating soil infiltration in two mix-reseeded plant communities via its developed fine roots.

Nevertheless, legumes are still crucial components of nitrogen-limited grassland ecosystems, and they make ideal forage for livestock because of high foliage protein content (Zhou et al., 2017). Therefore, we still recommend reseeded legumes together with grasses for restoration of a degraded grassland, since their mixtures tended to produce more forage yield for livestock than monocultures in this study (Table. S1). Ghani et al. (2022) also found that leguminous-gramineous mixtures (Alfalfa $\times$ S. brome grass and Alfalfa $\times$ T. fescue) significantly improved forage productivity over monocultures. Additionally, leguminous-gramineous communities could also provide more plant diversity than monocultures, which may be a great benefit for maintaining sustainability of grassland ecosystem (Pavlu et al., 2021; Zhang et al., 2018). Nevertheless, this study indicated that the *A. mongolicum* should be a necessary species selected with two legumes in mixtures for local grassland restoration, since *A. mongolicum* has more advantages than two legumes in regulating soil infiltrability.

In all, this study found reseeded native plants differently affected soil infiltration in local desert steppe, and the reseeded plants with high potential in regulating soil infiltrability may also benefit restoration of degraded grassland. This study also revealed that both plant root traits and soil properties significantly influenced soil infiltration in this reseeded desert steppe as other studies reported (Ferreira et al., 2010; Leung et al., 2015; Liu et al., 2015; Cui et al., 2019; Liu et al., 2020; Wu et al., 2021), yet we found root morphological characteristics in surface soil played a key role in determining soil infiltration. However, we highlight that both root morphological characteristics and soil properties were significantly determined by plant fine root volume rather than total root volume or/and total root biomass as others' reports (Liu et al., 2015; Huang et al., 2019; Cui et al., 2019; Liu et al., 2020). In addition, though some studies also found that fine root volume significantly influenced soil infiltration via improving root trait (Zhang et al., 2015) or via improving soil property (Cui et al., 2019), our study emphasize that future study should fully consider the contribution of fine root on improvements both for root trait and soil properties when soil infiltration will be concerned in arid and semi-arid areas.

5. Conclusions

In this study, all reseeded treatments benefit restoration of degraded grassland due to they increased soil infiltrability of local desert steppe. Moreover, soil infiltration was significantly influenced both by plant root morphological traits and soil porosities, yet root morphological

traits (root surface area and the root length density) in surface soil played a key role in determining soil infiltration. Nevertheless, both root morphological traits and soil porosities were significantly determined by volume of fine root. Therefore, this study indicated that the reseeded plant species, such as *A. mongolicum* in local desert steppe, with high fine root volume may benefit restoration of degraded grassland by acting to promote vegetation resilience and improve soil physical properties. Whereas, the current study is a case study, and future studies are still needed in other systems and locations to further explore the effects of fine roots on soil infiltration.

CRedit authorship contribution statement

Bo Wang: Investigation, Data curation, Writing – original draft. **Jing Liu:** Investigation, Data curation, Writing – original draft. **Zhigang Li:** Conceptualization, Methodology, Supervision, Funding acquisition, Writing – original draft, Writing – review & editing. **Stephen J. Morreale:** Writing – original draft, Writing – review & editing. **Rebecca L. Schneider:** Writing – original draft, Writing – review & editing. **Dongmei Xu:** Conceptualization, Methodology, Supervision, Funding acquisition. **Xiongkui Lin:** Investigation, Data curation, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This study was supported by the Key Research and Development Plan of Ningxia Hui Autonomous Region [2021BEG03010; 2021BEB04002], the Ningxia Natural Science Foundation [2022AAC05013], and the Ningxia Higher Education Institutions First-Class Discipline Construction Project [NXYLXK2017A01].

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.catena.2023.107020>.

References

- Alaoui, A., Caduff, U., Gerke, H.H., Weingartner, R., 2011. A preferential flow effects on infiltration and runoff in grassland and forest soils. *Vadose Zone J.* 10, 367–377. <https://doi.org/10.2136/vzj2010.0076>.
- Butler, P.J., Haygarth, P.M., 2007. Effects of tillage and reseeded on phosphorus transfers from grassland. *Soil Use Manag.* 23, 71–81. <https://doi.org/10.1111/j.1475-2743.2007.00107.x>.
- Cui, Z., Wu, G.L., Huang, Z., Liu, Y., 2019. Fine roots determine soil infiltration potential than soil water content in semi-arid grassland soils. *J. Hydrol.* 578, 124023 <https://doi.org/10.1016/j.jhydrol.2019.124023>.
- Cui, Z., Huang, Z., Luo, J., Qiu, K., López-Vicente, M., Wu, G.-L., 2021. Litter cover breaks soil water repellency of biocrusts, enhancing initial soil water infiltration and content in a semi-arid sandy land. *Agric Water Manag.* 255, 107009. <https://doi.org/10.1016/j.agwat.2021.107009>.
- Freschet, G.T., Roumet, C., 2017. Sampling roots to capture plant and soil functions. *Functional Ecology* 31, 1506–1518. <https://doi.org/10.1111/1365-2435.12883>.
- Galloway, A.F., Pedersen, M.J., Merry, B., Marcus, S.E., Blacker, J., Benning, L.G., Field, K.J., Knox, J.P., 2018. Xyloglucan is released by plants and promotes soil particle aggregation. *New Phytol.* 217, 1128–1136. <https://doi.org/10.1111/nph.14897>.
- Ghani, M.U., Kamran, M., Ahmad, I., Arshad, A., Zhang, C., Zhu, W., Lou, S., Hou, F., 2022. Alfalfa-grass mixtures reduce greenhouse gas emissions and net global warming potential while maintaining yield advantages over monocultures. *Sci. Total Environ.* 849, 157765 <https://doi.org/10.1016/j.scitotenv.2022.157765>.

- Guo, L., Liu, Y., Wu, G.-L., Huang, Z., Cui, Z., Cheng, Z., Zhang, R.-Q., Tian, F.-P., He, H., 2019. Preferential water flow: Influence of alfalfa (*Medicago sativa* L.) decayed root channels on soil water infiltration. *J. Hydrol.* 578, 124019. <https://doi.org/10.1016/j.jhydrol.2019.124019>.
- Guo, F.-X., Wang, Y.-P., Hou, T.-T., Zhang, L.-S., Mu, Y., Wu, F., 2021. Variation of soil moisture and fine roots distribution adopts rainwater collection, infiltration promoting and soil anti-seepage system (RCIP-SA) in hilly apple orchard on the Loess Plateau of China. *Agric Water Manag* 244, 106573. <https://doi.org/10.1016/j.agwat.2020.106573>.
- Hao, H., Wei, Y., Cao, D., Guo, Z., Shi, Z., 2020. Vegetation restoration and fine roots promote soil infiltrability in heavy-textured soils. *Soil Tillage Res.* 198, 104542. <https://doi.org/10.1016/j.still.2019.104542>.
- Huang, Z., Tian, F.P., Wu, G.L., Liu, Y., Dang, Z.Q., 2017. Legume grasslands promote precipitation infiltration better than gramineous grasslands in arid regions. *Land Degrad. Dev.* 28, 309–316. <https://onlinelibrary.wiley.com/doi/10.1002/ldr.2635>.
- Huang, Z., Sun, L., Liu, Y., Liu, Y.-F., López-Vicente, M., Wei, X.-H., Wu, G.-L., 2019. Alfalfa planting significantly improved alpine soil water infiltrability in the Qinghai-Tibetan Plateau. *Agr Ecosyst Environ* 285, 106606. <https://doi.org/10.1016/j.agee.2019.106606>.
- Ji, B., He, J., Wang, Z., Liu, F., Tian, B., Wu, X., Yu, H., Ren, X., Jiang, Q., 2022. Effects of tillage on soil carbon and nitrogen reserves in desert steppe of Ningxia. *Chinese J. Grassland* 44, 30–38. <https://doi.org/10.16742/j.zgdx.20210002>.
- Jotisankasa, A., Sirirattanachai, M.T., 2017. Effects of grass roots on soil-water retention curve and permeability. *Can. Geotech. J.* 54, 1612–1622. <https://doi.org/10.1139/CJ-2016-0281>.
- Kamran, M., Yan, Z., Chang, S., Chen, X., Ahmad, I., Jia, Q., Ghani, M.U., Nouman, M., Hou, F., 2022. Enhancing resource use efficiency of alfalfa with appropriate irrigation and fertilization strategy mitigate greenhouse gases emissions in the arid region of Northwest China. *Field Crop Res* 289, 108715. <https://doi.org/10.1016/j.fcr.2022.108715>.
- Le Bissonnais, Y., Prieto, I., Roumet, C., Nespoulous, J., Metayer, J., Huon, S., Villatoro, M., Stokes, A., 2018. Soil aggregate stability in Mediterranean and tropical agroecosystems: effect of plant roots and soil characteristics. *Plant and Soil* 424, 303–317. <https://doi.org/10.1007/s11104-017-3423-6>.
- Leung, A.K., Garg, A., Coe, J.L., Ng, C.W.W., Hau, B.C.H., 2015. Effects of the roots of *Cynodon dactylon* and *Schefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. *Hydrol. Process.* 29, 3342–3354. <https://doi.org/10.1002/hyp.10452>.
- Li, Y., Xu, D., Sa, C., Wang, X., Song, K., 2022. The study on improvement effect of native species on degraded desert steppe. *Chinese Journal of Grassland* 44, 36–45. <https://doi.org/10.16742/j.zgdx.20210236>.
- Liu, B., Xie, G., Zhang, X., Zhao, Y., Yin, X., Cheng, C., 2015. Vegetation root system, soil erosion and ecohydrology system: A review, in: *Proceedings of the 2015 International Forum on Energy, Environment Science and Materials*. Presented at the 2015 International Forum on Energy, Environment Science and Materials, Atlantis Press, Shenzhen, China. <https://doi.org/10.2991/ifeesm-15.2015.52>.
- Liu, Y., Cui, Z., Huang, Z., López-Vicente, M., Wu, G.-L., 2019. Influence of soil moisture and plant roots on the soil infiltration capacity at different stages in arid grasslands of China. *Catena* 182, 104147. <https://doi.org/10.1016/j.catena.2019.104147>.
- Liu, Y., Guo, L., Huang, Z., López-Vicente, M., Wu, G.-L., 2020. Root morphological characteristics and soil water infiltration capacity in semi-arid artificial grassland soils. *Agric Water Manag* 235, 106153. <https://doi.org/10.1016/j.agwat.2020.106153>.
- Liu, K., Sollenberger, L.E., Newman, Y.C., Vendramini, J.M.B., Interrante, S.M., White-Leach, R., 2011. Grazing management effects on productivity, nutritive value, and persistence of 'Tifton 85' Bermudagrass. *Crop Sci.* 51, 353–360. <https://doi.org/10.2135/cropsci2010.02.0122>.
- Liu, X., Zhang, G., Heathman, G.C., Wang, Y., Huang, C., 2009. Fractal features of soil particle-size distribution as affected by plant communities in the forested region of Mountain Yimeng, China. *Geoderma* 154, 123–130. <https://doi.org/10.1016/j.geoderma.2009.10.005>.
- Liu, G.X., Zhang, Y.J., Hovstad, K.A., Mao, P.S., Han, J.G., 2014. Competition of *Leymus chinensis* and *Bromus inermis* in response to gap size and neighbouring root exclusion. *Grass Forage Sci.* 69, 479–487. <https://doi.org/10.1111/gfs.12050>.
- Luo, Z., Niu, J., Xie, B., Zhang, L., Chen, X., Berndtsson, R., Du, J., Ao, J., Yang, L., Zhu, S., 2019. Influence of root distribution on preferential flow in deciduous and coniferous forest soils. *Forests* 10, 986. <https://doi.org/10.3390/f10110986>.
- Necpálová, M., Casey, I., Humphreys, J., 2013. Effect of ploughing and reseeded of permanent grassland on soil N, N leaching and nitrous oxide emissions from a clay-loam soil. *Nutrient Cycling Agroecosystems* 95, 305–317. <https://doi.org/10.1007/s10705-013-9564-y>.
- Nelson, D.W., Sommers, L.E., 1982. *Total Carbon, Organic Carbon, and Organic Matter*. In: *Page A.L. Methods of Soil Analysis. Part 2. Chemical and Microbiological Properties*. American Society of Agronomy and Soil Science Society of America, Madison, 9, 539–577.
- Neris, J., Jiménez, C., Fuentes, J., Morillas, G., Tejedor, M., 2012. Vegetation and land-use effects on soil properties and water infiltration of Andisols in Tenerife (Canary Islands, Spain). *Catena* 98, 55–62. <https://doi.org/10.1016/j.catena.2012.06.006>.
- Pachauri, R.K., Meyer, L.A., 2014. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. *Climatic Change* 2014 Synthesis Report, IPCC, Geneva.
- Pavlu, L., Pavlu, V.V., Fraser, M.D., 2021. What is the effect of 19 years of restoration managements on soil and vegetation on formerly improved upland grassland? *Sci. Total Environ.* 755, 142469. <https://doi.org/10.1016/j.scitotenv.2020.142469>.
- Shi, S., Zhao, F., Ren, X., Meng, Z., Dang, X., Wu, X., 2022. Soil infiltration properties are affected by typical plant communities in a semi-arid desert grassland in China. *Water* 14, 3301. <https://doi.org/10.3390/w14203301>.
- Six, J., Bossuyt, H., Degryse, S., Denef, K., 2004. A history of research on the link between (micro) aggregates, soil biota, and soil organic matter dynamics. *Soil Tillage and Research* 79, 7–31. <https://doi.org/10.1016/j.still.2004.03.008>.
- Solly, E.F., Schöning, I., Boch, S., Kandeler, E., Marhan, S., Michalzik, B., Müller, J., Zscheischler, J., Trumbore, S.E., Schimpf, M., 2014. Factors controlling decomposition rates of fine root litter in temperate forests and grasslands. *Plant Soil* 382, 203–218. <https://doi.org/10.1007/s11104-014-2151-4>.
- Sun, D., Yang, H., Guan, D.X., Yang, M., Wu, J.B., Yuan, F.H., Jin, C.J., Wang, A.Z., Zhang, Y.S., 2018. The effects of land use change on soil infiltration capacity in China: A meta-analysis. *Sci. Total Environ.* 626, 1394–1401. <https://doi.org/10.1016/j.scitotenv.2018.01.104>.
- Wang, H., Zhu, X., Zakari, S., Chen, C., Liu, W., Jiang, X.-J., 2022. Assessing the Effects of Plant Roots on Soil Water Infiltration Using Dyes and Hydrus-1D. *Forests* 13, 1095. <https://doi.org/10.3390/f13071095>.
- Wu, G.-L., Cui, Z., Huang, Z., 2021. Contribution of root decay process on soil infiltration capacity and soil water replenishment of planted forestland in semi-arid regions. *Geoderma* 404, 115289. <https://doi.org/10.1016/j.geoderma.2021.115289>.
- Zhang, Q., Liu, K., Shao, X., Li, H., He, Y., Sirimuji, W., B., 2021. Microbes require a relatively long time to recover in natural succession restoration of degraded grassland ecosystems. *Ecol. Ind.* 129, 107881. <https://doi.org/10.1016/j.ecolind.2021.107881>.
- Zhang, Y., Niu, J., Yu, X., Zhu, W., Du, X., 2015. Effects of fine root length density and root biomass on soil preferential flow in forest ecosystems. *Forest Systems* 24, e012.
- Zhang, Y., Lu, W., Zhang, H., Zhou, J., Shen, Y., 2018. Grassland management practices in Chinese steppes impact productivity, diversity and the relationship. *Front. Agric. Sci. Eng.* 5, 57. <https://doi.org/10.15302/J-FASE-2017192>.
- Zhang, Y.J., Zhou, J.Q., Yang, G.W., Jin, J.Y., Liu, N., 2020. Theory and application of no-tillage reseeded in degraded grasslands. *Chin. Sci. Bull.* 65, 1546–1555. <https://engine.scichina.com/doi/10.1360/TB-2020-0224>.
- Zhao, Y.G., Wu, P.T., Zhao, S.W., Feng, H., 2013. Variation of soil infiltrability across a 79-year chronosequence of naturally restored grassland on the loess plateau, China. *J. Hydrol.* 504, 94–103. <https://doi.org/10.1016/j.jhydrol.2013.09.039>.
- Zhou, J., Zhang, Y., Wilson, G.W.T., Cobb, A.B., Lu, W., Guo, Y., 2017. Small vegetation gaps increase reseeded yellow-flowered alfalfa performance and production in native grasslands. *Basic Appl. Ecol.* 24, 41–52. <https://doi.org/10.1016/j.baee.2017.08.002>.
- Zhou, J., Zhang, F., Huo, Y., Wilson, G.W.T., Cobb, A.B., Xu, X., Xiong, X., Liu, L., Zhang, Y., 2019. Following legume establishment, microbial and chemical associations facilitate improved productivity in degraded grasslands. *Plant and Soil* 443, 273–292. <https://doi.org/10.1007/s11104-019-04169-9>.